Problem 27.2

A perpetuity pays 1,000 at the beginning of the first year. Each subsequent payment is increased by inflation. If inflation is assumed to be 4% per year, calculate the present value of this perpetuity using an annual effective interest rate of 10%.

Solution

Index the payments by $n = 0, 1, 2, \ldots$, so the payment at time n is $1{,}000(1.04)^n$. Discounting at an annual effective interest rate of 10%, the present value is the geometric series

$$\begin{aligned} PV &= \sum_{n=0}^{\infty} \frac{1{,}000(1.04)^n}{(1.10)^n} \\ &= 1{,}000 \sum_{n=0}^{\infty} \left(\frac{1.04}{1.10}\right)^n \\ &= \frac{1{,}000}{1 - \frac{1.04}{1.10}} \\ &= 18{,}333.3333\ldots. \end{aligned}$$

Therefore, the present value of the perpetuity is

$$\boxed{18{,}333.33}.$$